CPE 381 Classwork 07: Building Blocks for Continuous-Time Signals

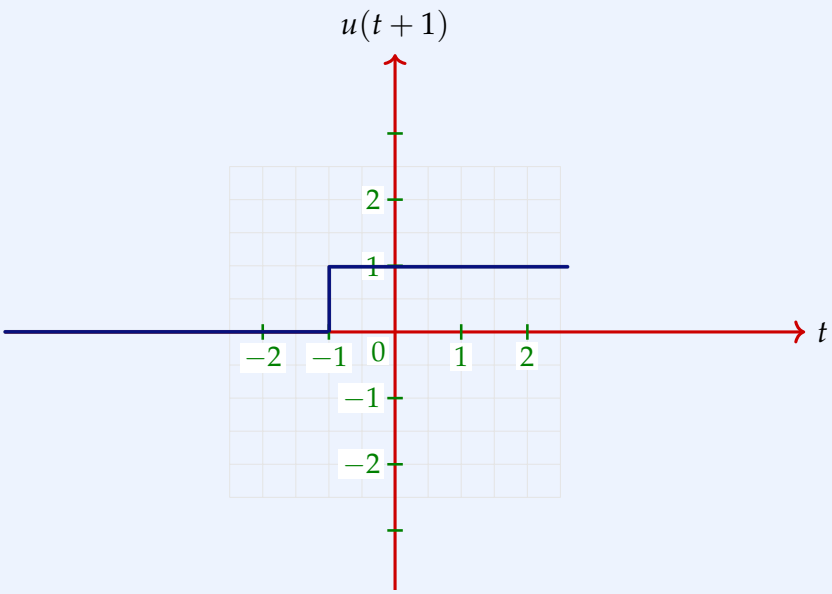
Student Name (as in Canvas): _____

A Number: _____

Points: 20 Duration: 15 minutes Due: September 16, 2026 (In-Class)

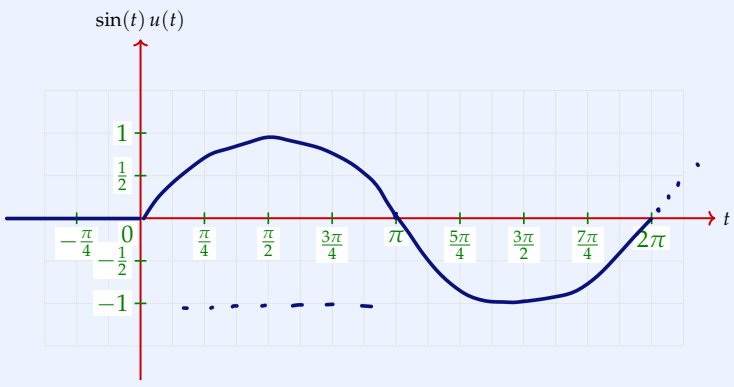
1. Assuming $u(t)$ is the unit-step function as discussed in the class, sketch $u(t+1)$.

Answer



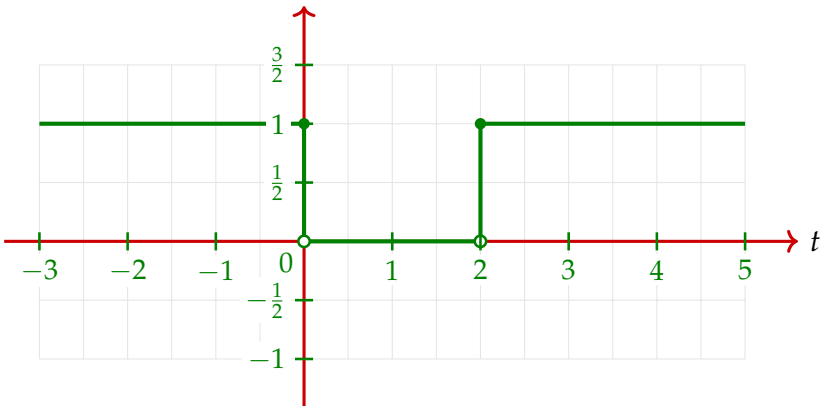
2. Sketch $x(t) = \sin(t)u(t)$.

Answer



3. Write mathematical equation corresponding to the following graph.

$u(t-2)+u(-t)$ ⟵ This is the solution.



Answer

$u(t-2)+u(-t)$

4. Calculate the energy of the signal

$$z(t)=\begin{cases}1, & 0\le t\le 10\\ 0, & \text{otherwise}\end{cases}$$

Answer

$$E_z=\int_{-\infty}^{\infty}|z(t)|^2=\int_{1}^{10}1^2\,dt$$

$$=(10-0)=10.$$